AIdeaLoan: "The Future of Agentic Credit Onboarding"

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AIdeaLoan

The Future of Agentic credit onboarding

From Compliance Friction → Conversational Sales

Fully Autonomous multimodal AI counselor
Replacing static video KYC

01.

ZERO-FORM INTERFACE

Natural, human-like video conversation replaces forms and clicks
→ Eliminates friction, reduces drop-offs, and boosts completion rates

02.

Physics-Based Security

Dynamic chromatic (color-based) prompts
Real-time liveness detection
→Detects light reflections & facial micro-expressions

03.

Global Accessibility

Real-time multilingual AI with seamless code-switching (100+ languages)
→ Expands reach to underserved and non-English-speaking users

The Dual Crisis: Friction vs. Fraud

Onboarding must be conversational to reduce friction and interactive to eliminate fraud

01.

The Onboarding Gap

- 60-70% of applicants abandon digital onboarding
- Driven by form fatigue & impersonal UX
- Billions lost in customer acquisition

02.

The Deepfake Threat

- Synthetic fraud projected to hit $40B annually by 2027
- Passive checks (blink, face scan) are now obsolete
- Generative AI can bypass traditional KYC defenses

03.

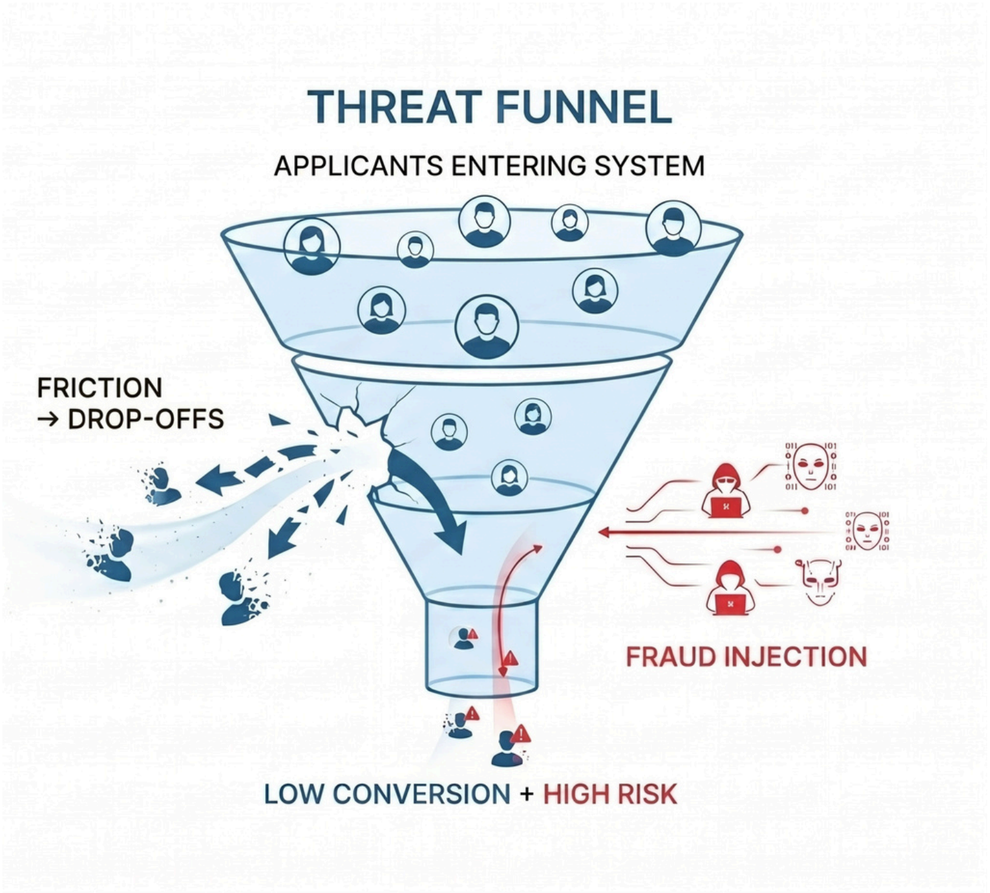
The Blindspot

- No real-time personalization during onboarding Driven by form fatigue & impersonal UX
- Missed upsell & engagement opportunities
- All users go through the same .

04.

The Compliance Bottleneck

- Manual review pipelines slow down scaling
- Inconsistent decisioning across cases
- Different reviewers → different outcomes → compliance risk



The Multimodal “ConvoGuardian” Agent

One conversation. Multiple intelligence streams. Instant decisions.

Agentic Multimodality

01

A unified LLM orchestrates speech + vision + reasoning in parallel
→ Combines STT/TTS, OCR, and biometric analysis in real-time

Conversational Logic

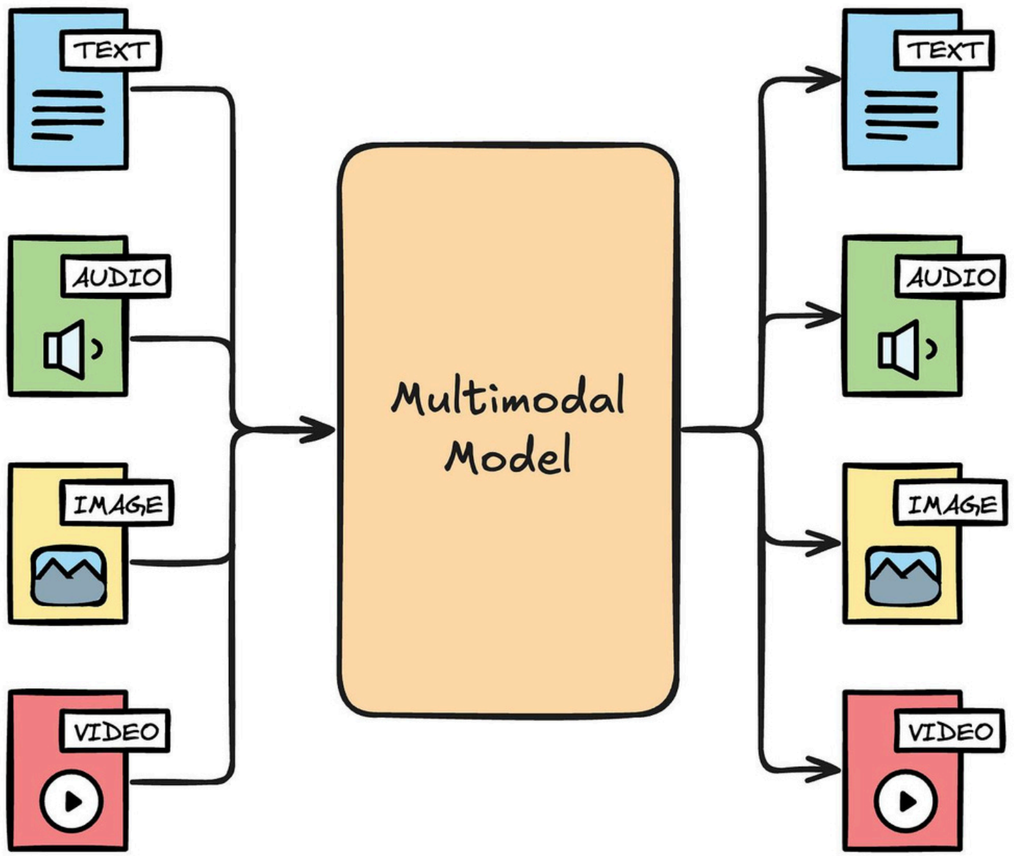
02

Users simply talk naturally (“I earn ₹60,000/month”)
→ AI extracts structured data, asks follow-ups, and guides document capture

Adaptive Intelligence

03

Context-aware, human-like interaction
→ Tone adapts dynamically + handles interruptions seamlessly



The Agentic Orchestrator: Real-Time Decision Engine

Frontend (Real-Time Interaction)

- WebRTC streaming (<1s latency)
- Zero-form interface (voice + video only)

Multimodal Pipelines

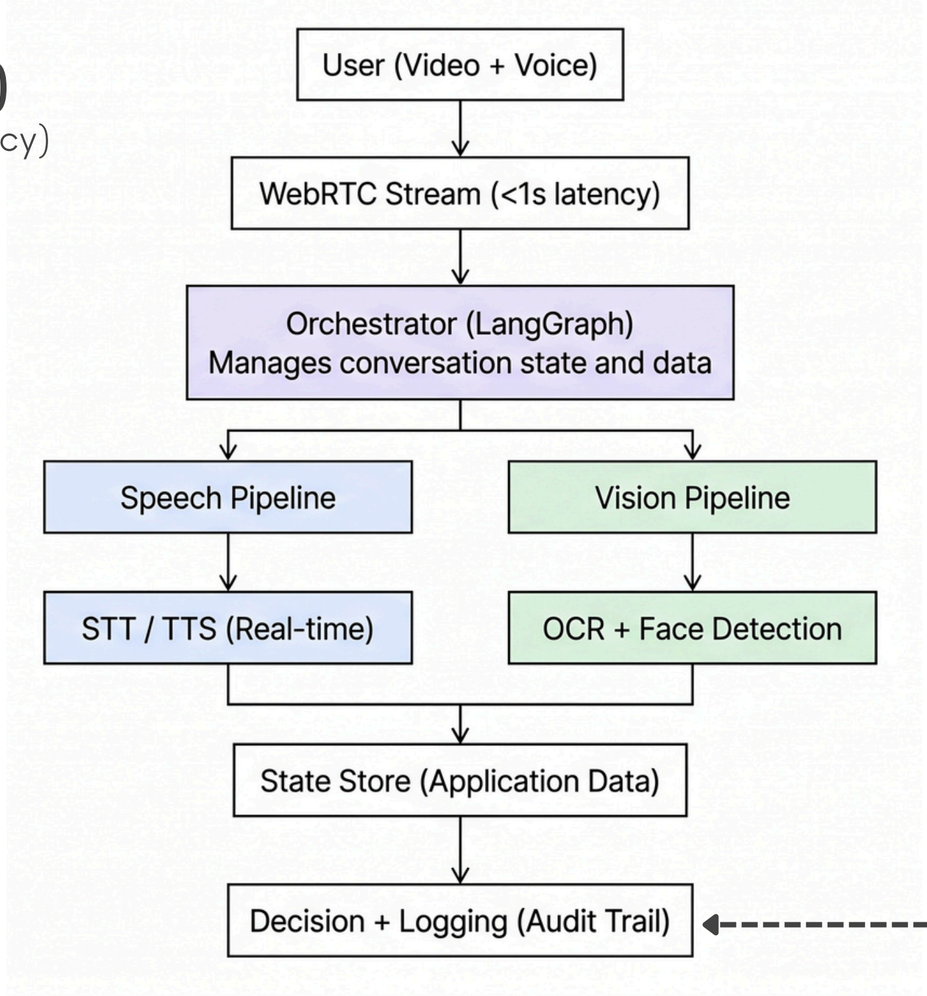
Speech

- Whisper STT
- Streaming TTS

Decision Layer (The Brain)

LLM dynamically:

- extracts user data
- identifies missing fields
- drives next question



Multimodal Pipelines

Vision

- MediaPipe (face + liveness cues)
- PaddleOCR (Documents)

Backend Integration

- Mock/real APIs: credit score, eligibility
- Compliance logging + audit trail

From raw speech + video → structured loan decision in real-time.

AldeaLoan

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Physics-Based Deepfake Detection

The Problem

- Deepfakes bypass blink / passive liveness
- Replay & injection attacks rising

Our Solution

- Random RGB light challenge (real-time)
- Captures micro-reflections on skin

Why it works

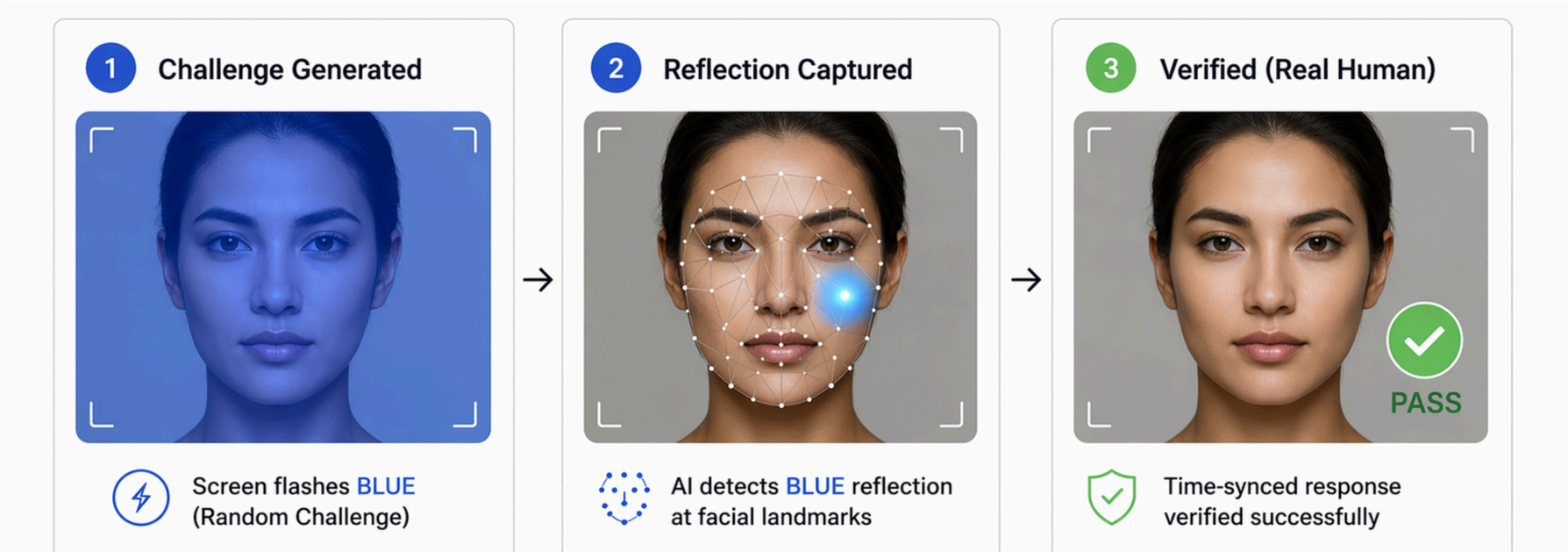
- Real skin → instant light reflection
- Deepfake / screen → delayed or inconsistent

Verification signal

- Time-synced color response at facial landmarks
- Impossible to pre-generate or replay

Performance

- ~98.8% spoof detection accuracy
- Works against replay + injection attacks



Integrity & Resilience: Auditor Agent & Adaptive Logic

VOICE TRANSCRIPT (ASR)

I earn ₹1,00,000

Confidence: High

AUDITOR AGENT ALERT

Mismatch Detected

→ Clarifying...

WHY FLAGGED?

Discrepancy between spoken value and document value

OCR DATA (PAYSLIP)

Monthly Payslip	
Employee Name:	John Doe
Employee ID:	JD12345
Income:	**₹80,000**
Net Pay:	₹80,000

Confidence: Medium

EXAMPLE User (Verbal): "Income = ₹1,0,00,000" → OCR (Payslip): "₹80K" → **System Flags + Asks Instantly**

Proactive Recovery → Flow

Call Drops — "Connection lost"

→ **Resume Link Sent** — "Instant link via SMS / WhatsApp"

→ **User Clicks Link** — "One-click rejoin"

→ **Session Resumes** — "Exact state restored"

→ **Continue Seamlessly** — "No data loss, no restart"

"Even if the user drops, the system doesn't."

Auditor Agent

Real-time Truth Engine

- Cross validates: Speech vs OCR vs Face
- Detects inconsistencies in real-time
- Triggers clarification prompts
- Ensures decsion integrity

Network Resilence

Adaptive to Poor Connectivity

- Dynamically adjusts stream: Video → Low bitrate → Audio fallback
- Designed for real-world Indian connectivity conditions

Proactive Recovery

Never Lose the Session

- Session state preserved continuously
- Instant recovery via: 1-click resume link (WhatsApp/SMS)
- Cross-device QR handoff
- User resumes exactly where they left

Behavioral Engagement & Governance Architecture

We aren't just checking boxes; we are managing the user's cognitive load and ensuring the "paper trail" is bulletproof for regulators.

Gamified Progression

- **Isometric Visualization:** Implements a Three.js **"Financial Bridge"** that renders structural segments in real-time as data is ingested.
- **Behavioral Impact:** Neutralizes the 60% industry abandonment rate by **replacing static forms** with **dopamine-driven visual** feedback loops.
- **Tech Stack:** React-ThreeFiber optimized for **low latency browser performance** on mobile devices.

Seamless Handoff

- **WebRTC State Sync:** Secure session migration via **tokenized QR codes,** allowing users to move from public kiosks to personal devices.
- **Conversational Persistence:** The LangGraph Orchestrator maintains session context in a c**entralized State Store**, ensuring zero dialogue reset during transitions.

Immutable Audit Trails

- **Compliance Logging:** Every verbal consent is captured as an encrypted, timestamped audio blob stored in a private ledger for **RBI forensics.**
- **Physics-Based Verification:** Integration of **Chromatic Liveness** results as cryptographic proof of physical presence, eliminating **deepfake injection risks.**

Summary and Impact

- **User Engagement:** Drastically reduces form fatigue and lowering abandonment rate.
- **Process Flow**: Frictionless kiosk transition to user personal device.
- **Compliance:** Bulletproof audit trail for regulators.

Market Displacement & Economic Moat

AldeaLoan isn't a utility; it's a profit center. We are moving from "Verification" to "Conversational Closing."

1. The ROI Breakdown: Quantifying Efficiency

- **45% Drop-off Recovery:** Achieved through Proactive Nudge Engines. If a session lags, the agent doesn't just wait; it uses **empathic tone-shifting** (e.g., "Take your time with that payslip, I'm right here if you have questions") to keep the user engaged.
- **80% OpEx Reduction**: Eliminates the need for massive BPO call centers. A single GPU-server can handle **100+ concurrent V-KYC sessions**, whereas a human officer can only handle one.

2. The Competitive "Conversational Alpha"

- **Legacy Players (Onfido, Cashfree, Jumio):** These are **"Static Checkers."** They are passive. If a user uploads a blurry ID, the system fails.
- **AldeaLoan:** It is an **Agentic Closer**. If a document is blurry, the agent speaks to the user in real-time: "Hey Tanay, that photo is a bit dark - can you tilt the card toward the light?"
- **Moat:** Our integration of parallel Speech/Vision pipelines creates a barrier to entry that simple wrapper-apps cannot replicate.

3. Scaling Vision: The "Next Billion" Users

- **Global Accessibility:** Real-time code-switching across **100+ languages** and **local dialects.**
- **Infrastructure Resilience:** Adaptive bitrate logic ensures the conversation continues even on **patchy 3G networks** in rural areas.
- **The Goal:** Turning the world's most **boring task (loan forms)** into a **5-minute natural conversation.**

Thank You